JAVA ENTERPRISE PRACTICALS

Practical 3: Cookie-Based Visit Counter Using Servlet

Environment: Apache NetBeans IDE · GlassFish Server · JDK 21

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1. Aim

To create a Java Servlet that uses cookies to store and display the number of times a user has visited the Servlet.

2. Software Requirements

- Java JDK 21
- Apache NetBeans IDE
- GlassFish Server
- Web browser

3. Theory

A cookie is a small piece of data stored in a web browser. A Servlet can create a cookie and send it to the browser. When the browser makes another matching request, it sends the cookie back to the Servlet.

In this practical, a cookie named `visitCount` stores the number of visits.

- **First visit:** count is 1.
- **Second visit:** count is 2.
- **Third visit:** count is 3.

The cookie is configured to remain valid for one day.

4. Algorithm

1. Create a Servlet named `VisitCounterServlet`.
2. Read cookies using `request.getCookies()`.
3. Search for the cookie named `visitCount`.
4. If the cookie is absent, initialize the count to 1.
5. If it exists, read the saved count and increment it.
6. Create a cookie with the updated count.
7. Add the cookie to the response.
8. Display the count in the browser.

5. Project Creation in NetBeans

1. Open Apache NetBeans.
2. Select **File** → **New Project** → **Java Web** → **Web Application**.
3. Enter project name: **CookieVisitCounter**.
4. Select GlassFish Server.
5. Choose the Java EE/Jakarta EE version supported by the server.
6. Click **Finish**.
7. Under Source Packages, create the package `com.cookies`.
8. Inside the package, create a Servlet named `VisitCounterServlet`.

6. Complete Java Program

File name: `VisitCounterServlet.java`
Package: `com.cookies`

The following code uses `jakarta.servlet`, suitable for GlassFish 6 or later.

```
package com.cookies;

import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

@WebServlet("/visit")
public class VisitCounterServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest request,
                         HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        int count = 1;

        // Read cookies sent by the browser
        Cookie[] cookies = request.getCookies();

        if (cookies != null) {
            for (Cookie cookie : cookies) {
                if ("visitCount".equals(cookie.getName())) {
                    try {
                        count = Integer.parseInt(
                                cookie.getValue()) + 1;
                    } catch (NumberFormatException e) {
                        count = 1;
                    }
                    break;
                }
            }
        }

        // Create a cookie with the updated visit count
        Cookie visitCookie = new Cookie(
                "visitCount", String.valueOf(count));

        // Keep the cookie for one day
        visitCookie.setMaxAge(24 * 60 * 60);

        // Set the cookie path
        visitCookie.setPath(
                request.getContextPath().isEmpty()
                ? "/" : request.getContextPath());

        // Send the cookie to the browser
        response.addCookie(visitCookie);

        // Display the visit count
        PrintWriter out = response.getWriter();
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head><title>Visit Counter</title></head>");
        out.println("<body>");
        out.println("<h2>Cookie-Based Visit Counter</h2>");
        out.println("<h3>You have visited this Servlet "
```

```
                + count + " time(s).</h3>");
        out.println("<p>Refresh the page to increase the count.</p>");
        out.println("</body>");
        out.println("</html>");
    }
}
```

Compatibility note: GlassFish 5 uses `javax.servlet` imports. GlassFish 6 and later use `jakarta.servlet` imports. Use the namespace supported by your server.

7. Important Code Explanation

Statement	Purpose
@WebServlet("/visit")	Maps the Servlet to the /visit URL.
doGet()	Handles HTTP GET requests.
request.getCookies()	Reads cookies sent by the browser.
cookie.getName()	Returns the cookie name.
cookie.getValue()	Returns the saved visit count.
new Cookie()	Creates a cookie.
setMaxAge()	Sets the cookie's lifetime in seconds.
response.addCookie()	Sends the cookie to the browser.
Integer.parseInt()	Converts the saved count to an integer.

8. Execution Steps

1. Save the Java file using **Ctrl + S**.
2. Right-click the project and select **Clean and Build**.
3. Check the Output window for errors.
4. Right-click the project and select **Run**.
5. Wait for GlassFish to deploy the application.
6. Open the Servlet URL in your browser.

Typical URL:
`http://localhost:8080/CookieVisitCounter/visit`
The port and context path may differ on your system.

7. Refresh the page several times.
8. Observe the visit count increasing.

9. Expected Output

First visit:

```
Cookie-Based Visit Counter
You have visited this Servlet 1 time(s).
```

Second visit:

```
Cookie-Based Visit Counter
You have visited this Servlet 2 time(s).
```

Third visit:

```
Cookie-Based Visit Counter
You have visited this Servlet 3 time(s).
```

The count can restart at 1 if the cookie expires, is deleted, or is not sent by the browser.

10. Common Errors and Solutions

Error: package jakarta.servlet does not exist

Check the GlassFish version and the project's server configuration. Ensure the correct Servlet API is available.

Error: HTTP 404

Check the Servlet mapping /visit and the application context path.

Counter always starts at 1

Check whether cookies are enabled, whether the browser is sending the cookie, and whether the cookie path matches the Servlet URL.

GlassFish fails to start

Check the Java platform configured in NetBeans and the server logs.

11. Viva Questions and Answers

Q1. What is a cookie?

A cookie is a small piece of data stored in a web browser and sent with matching HTTP requests.

Q2. Why are cookies used in this practical?

They allow the Servlet to remember the previous visit count between requests.

Q3. What does request.getCookies() do?

It retrieves cookies sent by the browser.

Q4. What does response.addCookie() do?

It adds a cookie to the HTTP response for the browser to store.

Q5. What is setMaxAge()?

It sets the cookie lifetime in seconds.

Q6. What is the difference between cookies and sessions?

Cookies are stored on the client. Session data is stored on the server, with a session identifier commonly maintained in a cookie.

Q7. Why do we use doGet()?

Opening or refreshing the Servlet URL normally generates an HTTP GET request.

Q8. Where is the visit count stored?

In the browser's visitCount cookie.

Q9. Does the counter work across different browsers?

Normally, no. Each browser or browser profile has its own cookies.

Q10. What happens when the cookie is deleted?

The Servlet cannot find the saved count and starts at 1 again.

12. Result

The Cookie-Based Visit Counter was implemented using Java Servlet and cookies. The application stores the visit count in a browser cookie and displays the updated count on subsequent visits.

13. Quick Revision

Remember this sequence:

1. `request.getCookies()`
2. Find `visitCount`.
3. Read and increment the count.

4. Create a new `Cookie`.
5. Call `response.addCookie()`.
6. Display the updated count.